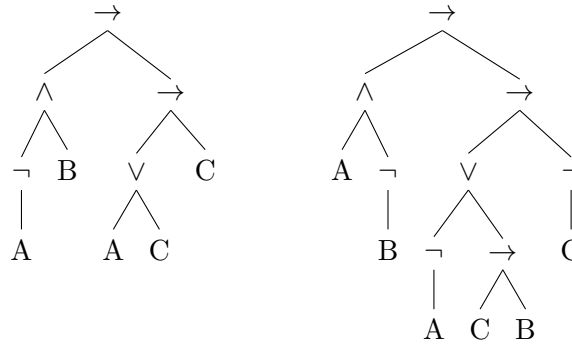


Exercise Sheet 12c - Solutions

Propositional Logic – Natural Deduction

1. The parse trees of F and G are



2. Here is a constructive Natural Deduction proof of $(\neg A \wedge B) \rightarrow (A \vee C) \rightarrow C$

$$\begin{array}{c}
 \overline{A}^3 \quad \overline{\neg A \wedge B}^1 \quad [\wedge E_L] \\
 \hline
 \overline{\neg A}^3 \quad [\neg E] \\
 \hline
 \perp \quad [\perp E] \\
 \hline
 \overline{A \vee C}^2 \quad \overline{A \rightarrow C}^3 \quad [\rightarrow I] \quad \overline{C \rightarrow C}^4 \quad [\rightarrow I] \\
 \hline
 \overline{C}^4 \quad [\vee E] \\
 \hline
 \overline{(A \vee C) \rightarrow C}^2 \quad [\rightarrow I] \\
 \hline
 \overline{(\neg A \wedge B) \rightarrow (A \vee C) \rightarrow C}^1 \quad [\rightarrow I]
 \end{array}$$

3. Here is a constructive Natural Deduction proof of $(A \wedge \neg B) \rightarrow (\neg A \vee (C \rightarrow B)) \rightarrow \neg C$

$$\begin{array}{c}
 \overline{A \wedge \neg B}^1 \quad [\wedge E_L] \quad \overline{A}^4 \quad [\neg E] \quad \overline{C \rightarrow B}^5 \quad \overline{C}^3 \quad [\rightarrow E] \quad \overline{A \wedge \neg B}^1 \quad [\wedge E_R] \\
 \hline
 \overline{A}^4 \quad [\neg E] \quad \overline{\neg B}^1 \quad [\neg E] \\
 \hline
 \overline{\neg A \vee (C \rightarrow B)}^2 \quad \overline{\perp}^4 \quad [\rightarrow I] \quad \overline{(C \rightarrow B) \rightarrow \perp}^5 \quad [\rightarrow I] \\
 \hline
 \overline{\perp}^3 \quad [\neg I] \\
 \hline
 \overline{\neg C}^3 \quad [\neg I] \\
 \hline
 \overline{(\neg A \vee (C \rightarrow B)) \rightarrow \neg C}^2 \quad [\rightarrow I] \\
 \hline
 \overline{(A \wedge \neg B) \rightarrow (\neg A \vee (C \rightarrow B)) \rightarrow \neg C}^1 \quad [\rightarrow I]
 \end{array}$$

4. F and G are provable and therefore valid by soundness, so any valuation satisfies the formula such as $A = \mathbf{T}$, $B = \mathbf{T}$, and $C = \mathbf{T}$.